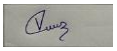
SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Karnesh V K / 192524296_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: 

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
   Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Question 1: Cloud Auto-Scaling and Load Balancing During Seasonal Sales

Question: A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Answer

1. Introduction

E-commerce companies often experience sudden increases in website traffic during seasonal sales, festival offers, and promotional events. During these periods, thousands or millions of users may access the website simultaneously. If the infrastructure cannot handle the increased demand, the website may become slow or unavailable. Cloud computing provides solutions such as auto-scaling, load balancing, and monitoring to manage these traffic fluctuations efficiently.

2. Cloud Auto-Scaling

Auto-scaling is a cloud computing feature that automatically increases or decreases computing resources according to application demand. When traffic increases, additional virtual machines or application instances can be launched. When demand decreases, unnecessary resources can be removed.

Benefits of auto-scaling:

- Automatically handles sudden increases in user traffic.
- Prevents server overload and improves application performance.
- Reduces manual intervention by administrators.
- Saves costs by releasing unused resources during low-demand periods.
- Maintains application availability during peak traffic.

3. Load Balancing

Load balancing distributes incoming user requests across multiple servers or application instances. Instead of sending all requests to a single server, the load balancer directs traffic to available servers according to their capacity and health.

Benefits of load balancing:

- Prevents individual servers from becoming overloaded.
- Distributes workloads efficiently.
- Improves response time for users.
- Redirects requests away from unhealthy servers.
- Supports high availability and fault tolerance.

4. Reducing Downtime and Improving User Experience

Auto-scaling and load balancing work together to maintain website performance during sudden traffic

spikes. Auto-scaling adds resources when demand increases, while load balancing distributes requests across the available resources.

If one server fails, the load balancer can redirect requests to healthy servers. This reduces the risk of complete service failure. Users experience faster page loading, smoother product searches, and more reliable checkout operations.

5. Role of Monitoring Tools

Cloud monitoring tools continuously track the performance and health of application infrastructure. They collect metrics such as CPU utilization, memory usage, request count, response time, and server errors.

Monitoring tools help administrators:

- Identify unusual traffic patterns.
- Detect performance bottlenecks.
- Configure auto-scaling policies based on resource utilization.
- Generate alerts when system performance decreases.
- Identify server failures and application errors.

6. Real-World E-Commerce Example

Consider an online shopping company conducting a festival sale. Normally, its website handles 10,000 users at a time, but during the sale, the number of users increases to 100,000.

The cloud platform automatically launches additional application instances. A load balancer distributes incoming requests among the available instances. Monitoring tools track server utilization and response time and trigger scaling actions when required.

As a result, customers can browse products, add items to their carts, and complete payments without significant delays. When the sale ends, the system reduces the number of instances to control costs.

Conclusion

Cloud auto-scaling and load balancing provide an efficient way to handle unpredictable traffic spikes. They improve scalability, availability, performance, and customer satisfaction while reducing infrastructure costs. Monitoring tools support this process by identifying problems and helping the cloud system respond quickly to changing demand.

Question 2: Secure Cloud Storage for Backup Data

Question: An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Answer

1. Introduction

Organizations generate large amounts of data, including customer records, business documents, financial information, and application databases. Maintaining backup copies is essential to protect this information from accidental deletion, hardware failure, cyberattacks, and natural disasters.

Cloud storage provides scalable and cost-effective backup facilities. It also supports data redundancy, encryption, access control, and disaster recovery.

2. Data Durability in Cloud Storage

Data durability refers to the ability of a storage system to preserve information and prevent permanent data loss over time. Cloud providers use reliable storage infrastructure, automated integrity checks, and redundant copies to protect stored data.

Important durability features include:

- Redundant storage: Multiple copies of data may be maintained across separate storage devices.
- Integrity checking: Checksums and other verification methods help detect data corruption.
- Fault-tolerant infrastructure: Storage systems are designed to continue operating when individual hardware components fail.
- Versioning: Previous versions of files can be retained so that users can recover earlier data.
- Automated backup: Backup schedules help ensure that important information is copied regularly.

The exact durability level depends on the storage service and its configuration.

3. Data Redundancy

Data redundancy means maintaining additional copies of information to reduce the risk of data loss.

Cloud storage can support different redundancy arrangements:

- Local redundancy: Data copies are stored on separate devices within a storage location.
- Zone redundancy: Data is replicated across separate availability zones, where supported.
- Geographical redundancy: Copies are maintained in different geographical locations or regions.

Geographical redundancy is particularly useful when a major disaster affects an entire data center or region.

4. Encryption and Access Control

Encryption protects data by converting readable information into an encoded form that can only be accessed using the appropriate decryption key.

Two important forms are:

- Encryption at rest: Protects backup files while they are stored.
- Encryption in transit: Protects data while it travels between the organization and cloud storage, usually through secure protocols such as TLS.

Access control ensures that only authorized users and applications can access backup information.

Security measures include:

- Role-based access control (RBAC).
- Multi-factor authentication (MFA).
- Least-privilege permissions.
- Secure key management.
- Audit logs that record access and administrative activities.
- Separate backup accounts or permissions to reduce the risk of attackers deleting backups.

These measures help protect sensitive data against unauthorized access, modification, and deletion.

5. Cost Advantages Compared with Traditional Storage

Traditional storage systems require organizations to purchase physical storage devices, servers, backup software, and supporting infrastructure. They must also provide electricity, cooling, maintenance, and physical security.

Cloud storage offers several cost advantages:

1. Pay-as-you-go pricing: Organizations pay according to the storage and services they use.
2. Reduced hardware investment: Large initial purchases of storage equipment may be avoided.
3. Scalability: Storage capacity can be increased without installing additional physical devices.
4. Automated management: Cloud services reduce the effort required for routine storage administration.
5. Storage tiers: Infrequently accessed backups can be moved to lower-cost archival storage, subject to retrieval charges and access delays.

However, organizations must consider data transfer, retrieval, replication, and long-term retention costs.

6. Disaster Recovery Example

Consider a hospital that stores patient records and medical images in its primary data center. A fire damages the data center, making the local systems unavailable.

The hospital maintains encrypted backups in cloud storage, with copies in a separate geographical region. After the disaster, the IT team restores the databases and application files from the cloud backups and reconnects them to the replacement infrastructure.

The hospital can recover important records and resume its services with less data loss and downtime than if it depended only on local backups.

A successful recovery also depends on regularly tested restoration procedures and clearly defined recovery time objectives (RTO) and recovery point objectives (RPO).

Conclusion

Cloud storage provides reliable backup facilities through redundancy, integrity checking, versioning, encryption, and access control. It reduces the need for extensive physical storage infrastructure and supports disaster recovery. By selecting appropriate storage tiers and security policies, organizations can protect critical information while controlling costs.

Question 3: Global Application Deployment Using CDNs and Edge Servers

Question: A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Answer

1. Introduction

When a mobile application serves users in different countries, the distance between users and application servers can affect response time. If every request must travel to a single data center, users located far away may experience delays, buffering, or poor performance.

Cloud computing helps companies deploy applications globally using content delivery networks (CDNs), edge servers, geographically distributed infrastructure, and traffic-routing services.

2. Content Delivery Network (CDN)

A Content Delivery Network is a distributed network of servers that delivers content to users from locations closer to them. It commonly caches static content such as images, videos, stylesheets, scripts, and downloadable files.

How a CDN works:

1. A user requests content through a mobile application.
2. The CDN directs the request to an appropriate edge location.
3. If the content is cached there, it is delivered directly to the user.
4. If the content is not available in the cache, the CDN retrieves it from the origin server and may cache it for future requests.

Benefits of CDNs:

- Reduce latency and loading time.
- Reduce repeated requests to the origin server.
- Lower bandwidth requirements at the origin.
- Improve content delivery during traffic spikes.
- Support users distributed across multiple geographical regions.

3. Role of Edge Servers

Edge servers are computing resources located closer to end users. They reduce the distance that requests and responses must travel.

Edge servers can improve application performance by:

- Caching frequently requested content.
- Processing certain requests near users.
- Reducing communication delays.

- Supporting regional traffic management.
- Providing faster responses for suitable applications.

However, not every task can be handled at the edge. Applications requiring centralized transactions or complex database operations may still need to communicate with backend systems.

4. Global Availability Through Cloud Providers

Major cloud providers offer infrastructure in multiple geographical regions and availability zones. Organizations can deploy application components in different locations to serve users closer to their locations.

Global availability can be supported through:

- Multi-region deployment: Application instances are deployed in multiple regions.
- Global traffic routing: User requests are directed to appropriate endpoints.
- Load balancing: Traffic is distributed across healthy application instances.
- Data replication: Data is copied between locations when supported by the database architecture.
- Health monitoring: Failed endpoints can be detected so that traffic is redirected.
- Disaster recovery: A secondary region can be used when the primary region becomes unavailable.

A globally distributed architecture must also consider data consistency, privacy regulations, and the additional cost of operating resources in multiple regions.

5. Real-Time Streaming Example

Consider a video-streaming application used by people in India, Europe, and North America. If all video files are delivered from a single server in India, viewers in other regions may experience buffering and slower playback.

By using a CDN, frequently watched video segments can be cached at edge locations near viewers. A user in Europe can receive cached video segments from a nearby edge server rather than repeatedly downloading them from India.

The application can also deploy its APIs and supporting services in multiple cloud regions. Load balancing and health monitoring help maintain service availability if a particular endpoint fails.

This architecture improves playback speed, reduces buffering, and supports a larger global audience.

Conclusion

CDNs and edge servers improve global application performance by delivering content closer to users and reducing unnecessary network delays. Cloud providers support global availability through multiple regions, load balancing, replication, and monitoring. These capabilities are especially useful for streaming, gaming, and other applications that require responsive services for users worldwide.

Question 4: Regulatory Compliance and Data Security in Cloud Computing

Question: A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Answer

1. Introduction

Financial institutions process sensitive information such as account details, payment records, customer identities, and transaction histories. Unauthorized access or data leakage can result in financial loss, legal penalties, and reduced customer trust.

When using cloud computing, financial organizations must protect their data and meet applicable regulatory requirements. Cloud providers offer security tools and compliance-related services, but the institution remains responsible for configuring its environment appropriately and meeting its own obligations.

2. Regulatory Compliance Support

Cloud providers maintain security controls, infrastructure protections, and compliance programs designed to support different industry and regional requirements.

Common forms of support include:

- Compliance reports and certifications for eligible cloud services.
- Identity and access management.
- Encryption and key-management services.
- Data retention and deletion controls.
- Security configuration tools.
- Audit logging and reporting.
- Documentation that helps customers assess their compliance responsibilities.

The organization must confirm that the specific cloud services and deployment arrangements meet the regulations applicable to its business. A provider's certification alone does not automatically make the customer's application compliant.

3. Role of Logging

Logging records events that occur within cloud infrastructure and applications. These records may include user logins, database access, administrative actions, configuration changes, and financial transaction events.

Logs help security teams:

- Identify unauthorized access attempts.
- Investigate suspicious activity.

- Trace changes made to critical resources.
- Support internal and external audits.
- Reconstruct the sequence of events after an incident.

Logs should be protected against unauthorized alteration and retained according to applicable policies and legal requirements.

4. Role of Monitoring

Monitoring continuously evaluates system activity and security-related metrics. It can identify unusual login patterns, unexpected data transfers, repeated failed authentication attempts, and abnormal application behavior.

Monitoring tools can generate alerts when predefined conditions occur. Security teams can then investigate the incident and respond quickly.

For example, if a customer account suddenly performs numerous high-value transfers from an unusual location, monitoring rules may alert the institution's fraud or security team.

Monitoring supports detection, but suspicious activity may still require investigation and additional fraud-prevention controls.

5. Role of Encryption

Encryption protects sensitive information from unauthorized disclosure.

- Encryption at rest: Protects stored customer records, transaction databases, and backups.
- Encryption in transit: Protects information moving between customers, applications, databases, and cloud services.
- Key management: Controls the creation, storage, use, rotation, and revocation of encryption keys.

For highly sensitive workloads, an institution may use customer-managed keys or hardware security modules where appropriate. Access to keys should be restricted and audited.

Encryption should be combined with strong authentication, least-privilege access, secure application design, and regular security assessments.

6. Ensuring Data Sovereignty

Data sovereignty refers to the legal and regulatory requirements associated with the country or jurisdiction where data is stored or processed.

A financial institution can address these requirements by:

1. Identifying the countries in which customer data may legally be stored and processed.
2. Selecting suitable cloud regions.
3. Configuring storage and database resources to use approved locations.
4. Reviewing where backups and replicated copies are maintained.
5. Examining cross-border access and data-transfer arrangements.
6. Checking contractual terms and relevant regulatory obligations.

7. Monitoring cloud configurations to prevent unauthorized location changes.

The institution should verify the location of backups, logs, support data, and replicated datasets instead of considering only the primary database location.

7. Example: Secure Financial Transactions

Consider a bank that processes online fund transfers through a cloud-hosted application. A customer initiates a transfer using the mobile banking application.

The system authenticates the customer and applies transaction authorization checks. Data is encrypted while travelling between the mobile application and the bank's services. Transaction records are stored in an appropriately protected database, with access restricted to authorized services and personnel. Logging records relevant transaction and administrative events, while monitoring detects suspicious patterns. The bank selects approved cloud regions for storing and processing regulated information and audits its configuration to ensure that data-location requirements are met.

This design helps the bank protect transaction information, investigate suspicious activity, and demonstrate its security controls during audits.

Conclusion

Cloud providers offer encryption, identity management, logging, monitoring, and compliance documentation that help financial institutions protect sensitive information. The institution must configure these controls correctly, review its legal responsibilities, and verify data-location requirements. Combining these measures supports secure transactions, effective auditing, and regulatory compliance.

Question 5: Containerization, Virtual Machines, Kubernetes, and Portability

Question: A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Answer

1. Introduction

Modern applications need to be deployed quickly, scaled according to demand, and operated consistently across development, testing, and production environments. Traditional deployment methods may create problems when software behaves differently on different servers.

Containerization packages an application with its required libraries and dependencies into a standardized unit called a container. Containers provide a consistent environment and make it easier to deploy applications across different infrastructure platforms.

2. Containers and Virtual Machines

A virtual machine (VM) emulates a complete computer system using a hypervisor. Each VM typically runs its own operating system, along with the applications installed inside it.

A container packages an application and its dependencies while sharing the host operating system's kernel. Containers are generally lighter than full virtual machines because they do not require a separate guest operating system for every application.

Feature	Containers	Virtual Machines
Architecture	Share the host OS kernel	Run separate guest operating systems
Startup time	Usually seconds or less, depending on the workload	Often takes longer because a guest OS must start
Resource usage	Generally lower overhead	Generally higher overhead
Isolation	Process and namespace-based isolation	Typically stronger OS-level separation through the hypervisor
Portability	Portable when the target environment supports the container runtime and architecture	Can run on compatible hypervisor platforms
Best suited for	Microservices and rapidly deployed applications	Full OS environments and workloads requiring VM-level isolation

Containers can improve deployment density and startup speed, but their security depends on correct configuration and isolation. Virtual machines remain useful when separate operating systems or stronger isolation boundaries are required.

3. Benefits of Containerization

Containerization provides several benefits:

- Consistency: Applications run with the same packaged dependencies across supported environments.
- Fast deployment: Containers can be started, stopped, and replaced quickly.
- Efficient resource usage: Multiple containers can share a host operating system.
- Simplified development: Developers can test applications in environments similar to production.
- Microservices support: Individual application services can be packaged and deployed independently.
- Version control: Container images can be tagged and maintained for different application releases.

These benefits help development teams deliver updates more reliably.

4. Container Orchestration Using Kubernetes

Kubernetes is a container orchestration platform that manages containerized applications across a cluster of machines.

Without orchestration, administrators may need to start containers manually, monitor their health, and handle failures individually. Kubernetes automates many of these tasks.

Its main functions include:

1. Automated deployment: Deploys containerized applications according to defined configurations.
2. Scaling: Increases or decreases the number of application replicas based on configured policies and metrics.
3. Self-healing: Restarts failed containers and replaces unhealthy application instances when configured to do so.
4. Service discovery: Provides stable service names and networking for communication between application components.
5. Load distribution: Distributes traffic among eligible service instances.
6. Rolling updates: Releases new application versions gradually.
7. Configuration management: Supplies application configuration and secrets through appropriate Kubernetes resources and security practices.

Kubernetes helps improve application availability and makes it easier to operate applications with many containers.

5. Portability Across Cloud Environments

Container images can be used across compatible environments, including local development systems, private cloud infrastructure, and public cloud platforms.

Portability benefits include:

- Reduced dependence on a single cloud provider's virtual-machine image format.
- Consistent application packaging across development and production.
- Easier migration between compatible environments.
- Support for hybrid-cloud and multi-cloud deployments.
- Simplified testing and deployment pipelines.

However, portability is not automatic in every case. Differences in networking, storage, identity management, managed databases, and provider-specific services may require configuration changes.

6. Microservices-Based Application Example

Consider an online shopping application consisting of several microservices:

- User service: Manages customer accounts and authentication.
- Product service: Stores product information and inventory details.
- Order service: Processes customer orders.

- Payment service: Integrates with payment systems.
- Notification service: Sends order confirmations and delivery updates.

Each service is packaged as a separate container image. Kubernetes deploys the containers and maintains the required replicas. If the product service receives more traffic than the notification service, the team can scale the product service independently.

If an instance fails, Kubernetes can replace it according to its health and deployment configuration. The same container images can also be deployed to another compatible cloud environment, subject to the required configuration and infrastructure being available.

This approach supports independent development, easier updates, and more flexible scaling.

Conclusion

Containers offer lightweight application packaging and generally use fewer resources than virtual machines. Kubernetes simplifies container deployment, scaling, health management, and service discovery. Container images also improve portability across compatible cloud environments. For microservices-based applications, containerization provides an efficient way to deploy, maintain, and scale individual services.